Project Idea Document

Trilingual Multimodal Financial Support Ticket Classification and Priority Prediction

1. Proposed Project Title

A Trilingual Multimodal Financial Support Ticket Triage System Using Multilingual Transformers and Existing Audio/Image Models.

2. Background and Motivation

Financial institutions receive support requests through multiple channels such as online forms, emails, chat messages, voice recordings, and screenshots of failed transactions. In Sri Lanka, users may communicate in English, Sinhala, Tamil, or code-mixed language. A manual ticket handling process can be slow, inconsistent, and difficult to scale when ticket volume increases.

This project proposes an AI-assisted triage system that classifies financial support tickets into predefined categories, predicts urgency, identifies customer sentiment, and uses audio/image attachments as additional evidence. The system is designed to assist support agents, not replace them.

3. Problem Statement

Existing support workflows often depend on human agents to read each ticket, understand the issue, assign a category, decide priority, and respond to the customer. This becomes harder when tickets are submitted in different languages or with incomplete information. Audio complaints and screenshots may contain important clues, but they are often not used in automated classification.

The research problem is to benchmark how effectively existing multilingual and multimodal AI models can support trilingual financial ticket triage across English, Sinhala, and Tamil.

4. Aim and Objectives

Aim: To design and evaluate a trilingual multimodal financial support ticket triage system for category classification, priority prediction, and sentiment classification.

- Build a dataset strategy using public financial complaint data, translated/adapted samples, and manually created Sinhala/Tamil/code-mixed examples.
- Benchmark traditional ML, multilingual transformer models, and LLM prompting for category, priority, and sentiment tasks.
- Add audio support by converting user voice complaints into text using an existing speech-to-text model, then classifying the transcript.
- Add image support by extracting text from transaction screenshots or receipts using OCR, then combining it with the user ticket text.
- Develop a simple web application and dashboard for ticket submission, prediction display, analytics, and human correction.

5. Proposed Scope

Component	Project Scope
Domain	Financial/banking support tickets: transactions, cards, online banking, fraud/security, loans, account/KYC, and general inquiries.
Languages	English, Sinhala, Tamil, and common code-mixed forms such as Sinhala-English or Tamil-English.
Inputs	Text complaint, optional audio recording, and optional image attachment such as receipt, bank slip, or failed transaction screenshot.
Core outputs	Predicted category, priority level, sentiment label, detected language, confidence score, and optional response suggestion.
Users	Customer submits ticket; support agent/admin reviews predictions and dashboard analytics.

6. Methodology and System Architecture

The system follows an extraction-then-classification design. Text tickets are processed directly. Audio tickets are first transcribed into text. Image attachments are processed through OCR to extract visible text such as transaction amount, error message, or receipt details. The extracted text is then combined and passed into the same trilingual classification pipeline.

Stage	Method
Text input	Preprocess text → language detection → classification models
Audio input	Speech-to-text → transcript cleaning → classification models
Image input	OCR/text extraction → evidence merging → classification models
Prediction layer	Category classifier + priority model + sentiment classifier
Dashboard	Ticket list, charts, confidence scores, and agent correction feedback

7. Research Benchmarking Plan

- Text benchmark: compare TF-IDF + SVM/Logistic Regression, mBERT, XLM-RoBERTa, Indic-language models, and LLM zero-shot/few-shot prompting.
- Translation benchmark: compare native multilingual classification against Sinhala/Tamil-to-English translation followed by an English classifier.
- Audio benchmark: compare classification accuracy on clean typed text versus speech-to-text transcripts.
- Image benchmark: compare text-only classification against text + OCR evidence from screenshots or receipts.
- Evaluation metrics: Accuracy, Precision, Recall, Macro-F1, confusion matrix, latency, and error analysis by language and input type.

8. Dataset Strategy

The main dataset can be built from public financial complaint datasets such as CFPB complaint data and banking intent datasets such as BANKING77. Since these sources are mainly English, a representative subset should be translated/adapted into Sinhala and Tamil, then manually verified. Additional synthetic and manually written tickets can be created for Sri Lankan banking scenarios and code-mixed usage.

A practical minimum dataset could contain 1,500–3,000 tickets across languages, balanced by category and priority. A larger benchmark can be created by selecting a larger English subset and translating a controlled portion into Sinhala and Tamil.

9. Expected Outputs and Contribution

- A working web application where users submit financial tickets using text, audio, or image attachments.
- A dashboard showing ticket category, priority, sentiment, language, confidence, and analytics.
- A benchmark report comparing multilingual models across English, Sinhala, Tamil, and multimodal input conditions.
- A clear research contribution on the effect of language, code-mixing, ASR errors, and OCR evidence on financial ticket triage.

10. Key References / Data Sources

- CFPB Consumer Complaint Database — official financial complaint dataset source.
- BANKING77 — English online banking intent classification dataset with 77 intents.
- Whisper — multilingual speech recognition model suitable for audio ticket transcription.
- Tesseract OCR — open-source OCR engine with multilingual support for image text extraction.